

Supplementary information for OpenDC-LCA

Han Hu^a, Darin W. Nutter^a

^a*Department of Mechanical Engineering, University of Arkansas, Fayetteville, Arkansas, USA*

This supplement documents secondary analyses, illustrative software cases, and reproducibility details that are useful for audit but not necessary to the main argument. It must be read with the evidence classifications in the manuscript.

S1. Evidence inventory and provenance

The repository source registry records each provider-native file, version, access URL, declared license or redistribution constraint, local filename, and SHA-256 checksum. Raw files with uncertain redistribution rights remain under `private-data/` and are excluded from Git. Derived, non-provider tables are written to both `paper/tables/` and a results directory.

The manuscript analysis uses the following provider fields:

- Microsoft/WSP: normalized GHG, primary-energy, and blue-water contribution tables; detailed GaBi electricity endpoints; pedigree assessment.
- eGRID: STNGENAN, STCO2AN, STCH4AN, STN20AN, and STC2ERTA at the state level and corresponding US* fields at the U.S. level.

- Boavizta: product category, manufacturer, product name, report date, lifetime, total GWP, and manufacturing share.
- ÖKOBAUDAT: UUID, product, geography, declared unit, modules A1-A3, GWP, nonrenewable primary energy, and freshwater.

USLCI, GLAD hydrogen, and USGS records are registered and exchange-tested but do not enter a reported LCIA total.

S2. Reproducibility table map

Table S1: Supplementary result map

File	Content	Evidence interpretation
table1_microsoft_reproduction_audit.csv	24 released totals and reconstruction error	Arithmetic consistency only
table2_grid_reductions_vs_air.csv	Relative reductions from the released model	Reconstructed released result
table6_state_rebased_ghg.csv	2023 state screening outputs	Boundary-mismatched screening transformation

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Table S1: Supplementary result map (continued)

File	Content	Evidence interpretation
table8_crossover_thresholds.csv	Pairwise crossover values	Conditional on a numerical transfer between the two released electricity endpoints
table9_boavizta_server_records.csv	48 public product records	Heterogeneous secondary evidence
table11_material_decarbonization_levers.csv	Selected matched process comparisons	Per-unit procurement leverage only
table14_egrid_historical_state_factors.csv	459 harmonized state-year rates	Direct generation-rate scenarios
table15_egrid_historical_national_factors.csv	Harmonized and provider U.S. series	Historical trend and scope audit
table21_anchor_extrapolation_diagnostic.csv	Within/below/above anchor counts	Transformation-validity diagnostic
table22_boundary_mismatch_stress.csv	Additive allowance cases	Sensitivity, not an upstream estimate

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Table S1: Supplementary result map (continued)

File	Content	Evidence interpretation
<code>table23_joint_assumption_stress.csv</code>	Seeded stress frequencies	Not probability or confidence
<code>table24_functional_unit_sensitivity.csv</code>	Impact-per-service break-even correction	Measurement-resolution target
<code>table25_priority_index_sensitivity.csv</code>	Rank stability across index weights	Heuristic diagnostic

S3. Released primary-energy and blue-water results

The 24-cell reconstruction includes GHG, primary energy, and blue water. The main manuscript focuses on GHG because the historical eGRID extension supplies gas-specific air-emission data, not a lifecycle primary-energy or electricity-mediated water inventory.

At the released U.S. grid endpoint, the Microsoft/WSP normalized results show that cold plate, single-phase immersion, and two-phase immersion reduce the three indicators relative to air cooling. These results inherit the released functional unit, boundaries, inventories, allocation, performance, and electricity cases. They were not re-based with eGRID for primary energy or water.

S4. Pairwise crossover screen

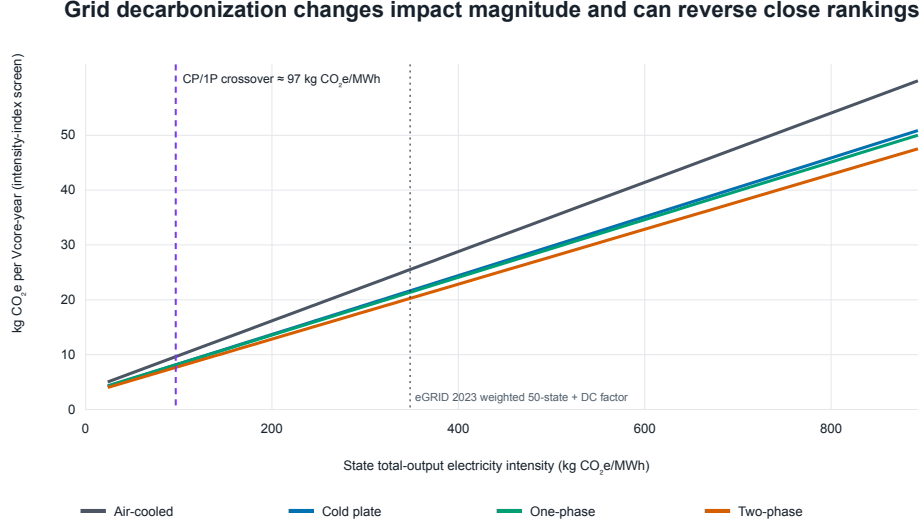


Figure S1: Screening totals transferred across 2023 eGRID state/DC rates. The vertical line is the cold-plate/single-phase crossover. The x-axis is a numerical electricity-intensity index across incompatible lifecycle boundaries, not a background-process substitution.

The two-phase architecture has no crossover with cold plate or single-phase inside the 2023 state-rate range under the fixed released foreground. This absence is conditional on server performance, fluid, equipment quantity, lifetime, and the electricity transfer model.

S5. Deterministic performance discrimination

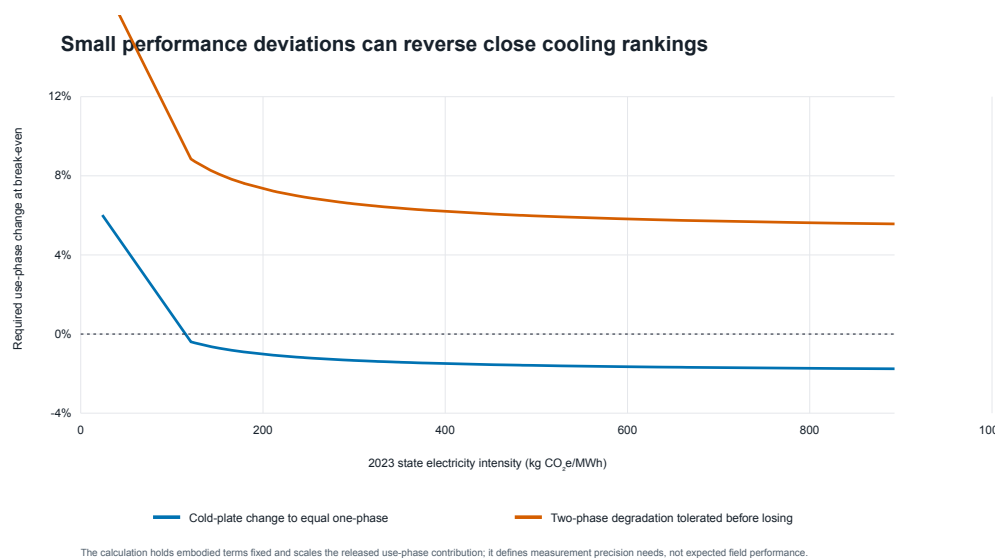


Figure S2: One-at-a-time use-phase change required to alter close rankings across 2023 state rates. These thresholds define experimental discrimination requirements and are not uncertainty intervals.

The cold-plate/single-phase order can be reversed by small use-phase changes. The two-phase margin is larger in the one-at-a-time test but becomes less decisive under joint stress, as reported in the main manuscript.

S6. Boavizta server-product dispersion

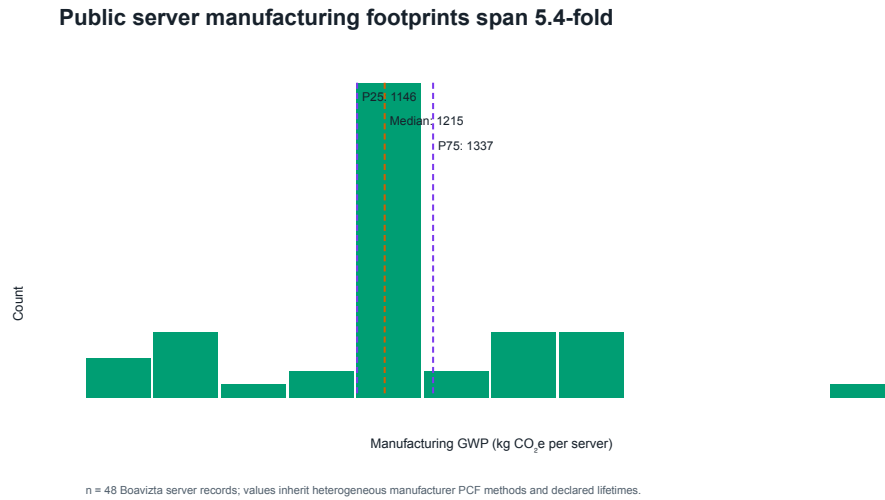


Figure S3: Distribution of manufacturing GHG across 48 Boavizta server records. The records inherit heterogeneous product rules, hardware configurations, lifetimes, and source methods.

The common-scaling stress applies the empirical annualized dispersion to the released compute, storage, and networking contributions simultaneously. It is not a product substitution model. Decision-grade use requires architecture-specific server count, configuration, useful computation, and lifetime under harmonized product-carbon-footprint rules.

S7. ÖKOBAUDAT material-route screen

Open construction datasets expose large procurement levers

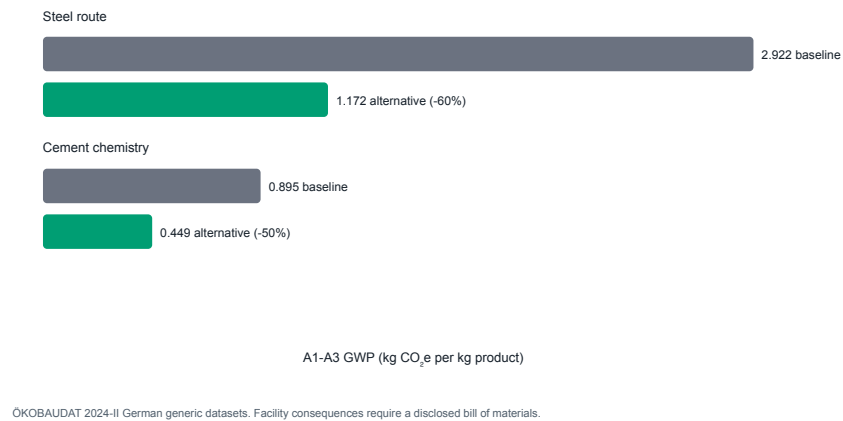


Figure S4: Selected matched ÖKOBAUDAT A1-A3 GHG factors. The comparison identifies per-kilogram procurement leverage and is not propagated to a data-center total.

The selected records are German generic datasets. A facility-level analysis requires architecture-specific quantities, material grade and supplier, fabrication, transport, replacement, and end-of-life alignment.

S8. Hourly climate and performance-map examples

The Fayetteville TMY file contains 8,760 hourly weather records. The repository uses it with a synthetic cooling-performance surface to test:

- schema and unit validation;
- bounded bilinear interpolation;

- uncovered-domain rejection;
- hourly PUE and operational-impact integration; and
- CLI, API, GUI, and report consistency.

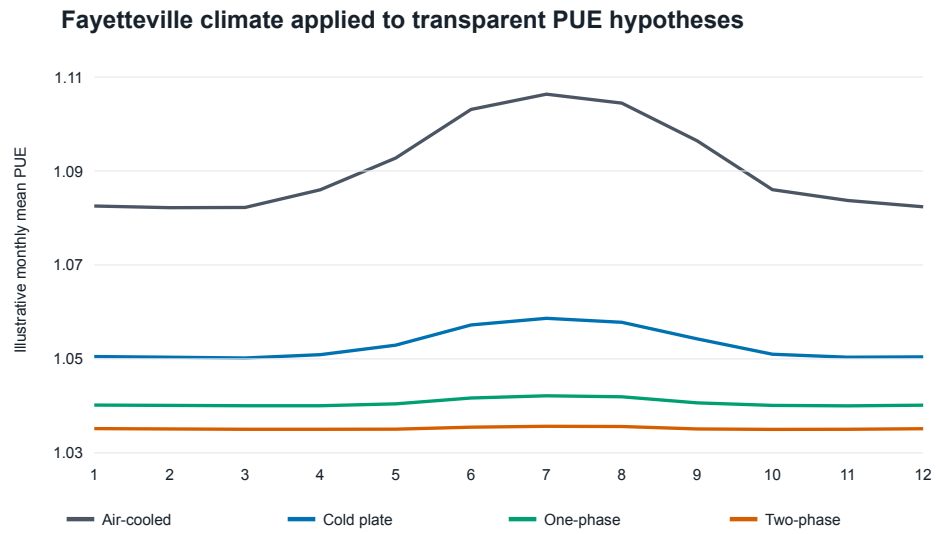


Figure S5: Monthly PUE from the Fayetteville weather and declared illustrative performance surface. This is a software example, not empirical evidence for a cooling technology.

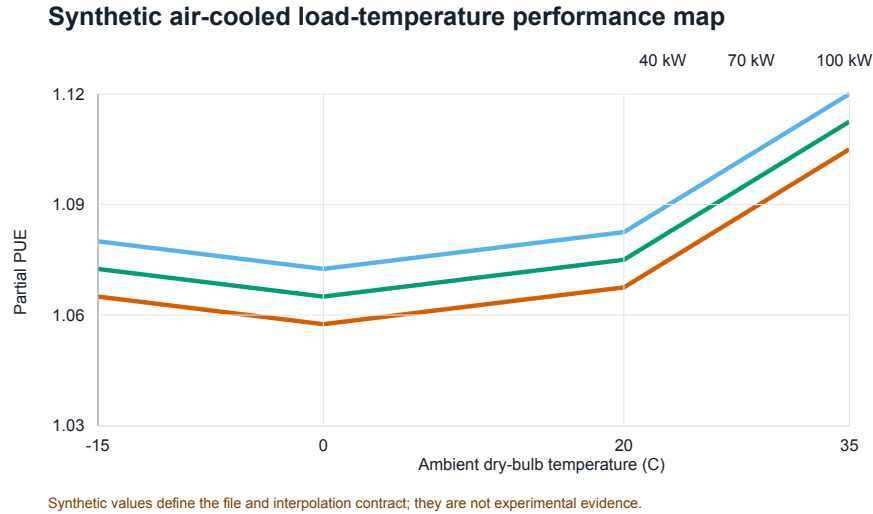


Figure S6: Synthetic load-temperature cooling surface used to verify bounded interpolation. No comparative assertion is permitted from this fixture.

Submission-grade energy results require architecture-specific measured power for IT-integral fans, pumps, coolant distribution, heat rejection, controls, and onsite water over a common load-weather domain.

S9. Data-quality priority and its limitations

The base priority index multiplies mean released GHG contribution by normalized pedigree weakness. Alternative exponents show that use phase is usually, but not invariably, first. Storage remains consistently in the top tier. This robustness test prevents the heuristic index from being described as a formal value-of-information result.

Formal value of information would additionally require:

- empirically defensible parameter distributions and correlations;

- the probability and consequence of selecting the wrong architecture;
- cost, duration, and achievable precision of each measurement or inventory;
- a decision threshold and stakeholder utility; and
- posterior updating after data acquisition.

S10. Reliability and replacement

The package supports discrete replacement, linearized replacement, Weibull renewal expectation, and optional Arrhenius acceleration. These functions are tested independently of the Microsoft/WSP secondary analysis. They do not provide architecture-specific reliability without failure and maintenance data.

Required records include component definition, population, duty cycle, temperature history, failure mode, censoring, repair/replacement action, downtime, spare inventory, fluid loss, and calibration/inspection history. Dependent failures, redundancy, and repair queues require an extended model.

S11. Interoperability boundary

Successful openLCA JSON-LD parsing or Brightway mapping establishes syntactic exchange. Numerical compatibility additionally requires matching:

- reference product and unit;
- provider and database version;

- geography and reference year;
- attributional or consequential system model;
- allocation and cut-off;
- elementary-flow nomenclature;
- LCIA method and version; and
- foreground product-system linkage.

The seven GLAD hydrogen archives and USLCI database are therefore preserved as future inputs, not silently substituted into the current cooling results.

S12. Reproduction commands

From the repository root:

```
python -m pip install -e ".[dev,gui]"
python scripts/run_integrated_evidence_analysis.py
python scripts/run_applied_energy_analysis.py
pytest -q
python paper/build_manuscript.py
python paper/build_overleaf.py
```

The scripts use fixed output paths, newline-stable CSV writing, and a seeded joint stress analysis. The raw-provider file manifest and derived tables should be archived with the exact Git commit used for submission.